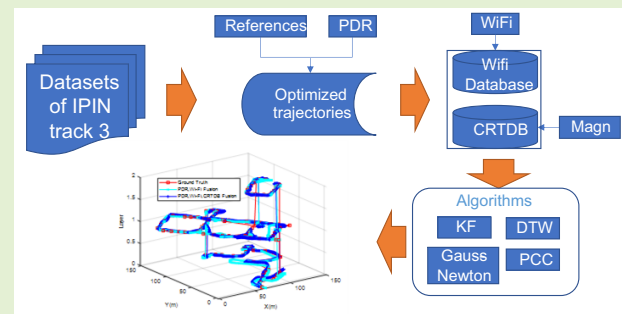


Indoor Positioning Based on Walking-Surveyed Wi-Fi Fingerprint and Corner Reference Trajectory-Geomagnetic Database

Yuan Wu^{1b}, Ruizhi Chen^{1b}, Wei Li, Yue Yu^{1b}, Haitao Zhou, and Ke Yan

Abstract—Indoor positioning has attracted many attentions in recent years. Wi-Fi, geomagnetic and inertial sensors are commonly used sources to reconstruct pedestrian's trajectory. But the error drift of inertial sensors and the labor-intensive fingerprints construction constrain their popularity. The purpose of this paper is to design a Wi-Fi, geomagnetic and pedestrian dead-reckoning (PDR) composed indoor positioning system which is accurate, practical and less labor-intensive. In order to achieve this goal, we design an indoor positioning system based on walking-surveyed Wi-Fi fingerprint and corner reference trajectory-geomagnetic database (CRTDB). Firstly, we propose a trajectory optimization algorithm which makes use of landmark observations to optimize historical positions of PDR by means of Gauss-Newton algorithm. Next, we construct the walking-surveyed Wi-Fi fingerprint database and the CRTDB using the optimized PDR trajectories. Secondly, we propose a Dynamic Time Warping (DTW) and Pearson Correlation Coefficient (PCC) combined CRTDB matching algorithm to find the coordinates of the observed corners. Thirdly, we propose a PDR, Wi-Fi fingerprint and CRTDB fused positioning system based on Kalman filtering (KF). Finally, we evaluate the performance and effectiveness of our proposed algorithms on two open datasets. The experimental results show that with our proposed CRTDB optimizing, the median error is improved by more than 33% and 27% respectively compared with that of PDR and Wi-Fi fusion.

Index Terms—Indoor positioning, Wi-Fi fingerprint database, PDR, corner reference trajectory-geomagnetic database (CRTDB), dynamic time warping (DTW), Pearson correlation coefficient (PCC), Gauss-Newton, Kalman filtering (KF).



I. INTRODUCTION

WITH the development of internet of things (IoT), the location-based service (LBS) is attracting more and more attentions of investors. In outdoor environment, Global Positioning System (GPS) has provided most of the positioning service. But there is lack of a universal positioning solution in indoor scenes where most of the human activities happened but the GPS signal is blocked. So far, a variety of indoor positioning techniques based on many kinds of sensors or signals have been proposed, e.g., Bluetooth [1], Wireless Fidelity (Wi-Fi) [2], radio frequency identifier (RFID) [3], Ultra-Wide Band (UWB) [4], Ultrasonic [5], light [6],

geomagnetic [7], inertial sensors [8] and camera [9]. These solutions can be divided into two categories: the infrastructure-based solution (IBS) (e.g., Bluetooth, Wi-Fi, RFID, UWB, Ultrasonic) and the infrastructure-free solution (IFS) (e.g., geomagnetic, inertial sensors, camera) [10]. IBS is characterized by the requirement of installing special devices in the buildings while IFS is only based on the carried sensors. Initial position is usually provided by IBS. A comprehensive overview of existing efforts on indoor positioning was discussed by Ali *et al.* in [11].

Wi-Fi positioning (WP) has attracted many attentions among all of the IBSs because of its universality. It doesn't need to install additional devices in the environment. Wi-Fi signals have covered most of the indoor spaces. And every smartphone has a Wi-Fi module [12]. Thus, it is able to build a positioning system without other hardware costs. However, the distribution of Wi-Fi signals will change with time and space. This will cause some specific problems, such as non-line-of-sight [13], labor-intensive database updating [14], etc.

Geomagnetic matching (GM) method uses the special distribution of the disturbed geomagnetic in indoors to recognize the position. The omnipresence of geomagnetic field makes it possible to design a self-contained positioning algorithm. But the point-by-point establishment of geomagnetic

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fingerprint (GF) is also labor-intensive. Hence the walk-survey based methods are adopted by some researchers, such as [15].

Generally, Wi-Fi fingerprinting is able to provide a global location accuracy while magnetic fingerprinting works only locally. The accuracy of Wi-Fi fingerprinting is relatively low. On the contrary, magnetic fingerprinting can achieve a higher accuracy. The combination of Wi-Fi fingerprints and magnetic fingerprints can compensate the drawbacks of two methods, achieving a high accuracy [16].

Inertial measurement unit (IMU), usually combining accelerometer, gyroscope and magnetometer, is now equipped on every smart-phone. IMU measurements can be used to solve the attitude relative to local coordinate and the distance change relative to start point, thereby reckoning the position relative to the initial point. This process can be realized by pedestrian dead-reckoning (PDR) or inertial navigation system (INS) [17]. In short time, PDR is able to provide a high accurate position estimation. However, disturbed by drift of IMU, the position estimated by PDR drifts rapidly with time [18]. PDR is generally only used to provide 2-D (Dimensional) trajectory. Hence, the height or layer of 3-D trajectory is usually estimated by other means, such as barometer.

The labor-intensive problem of fingerprint establishment can be alleviated to a certain extent by walk-survey based methods. And the drift of PDR can be solved by fusing other positioning methods. Hence, this research is to design a novel Wi-Fi, PDR and geomagnetic fused indoor positioning system which is accurate, self-contained, less labor-intensive and possible established by crowdsourcing ways. To realize this goal, we are going to propose a fast and accurate way to construct the Wi-Fi database with less site-surveying efforts. The accurate database construction needs accurate reference points as the landmarks of the radio signals of Wi-Fi. However, PDR cannot solely provide accurate location without correction of landmarks. Hence, we leverage the environment knowledge, the turn corners (left-turn, right-turn and U-turn), as the landmarks to revise the PDR trajectory. This needs the true coordinate of the turn corners which can be acquired on the building map or by pre-labeled means. The only need for the corner landmarks reduces the labor efforts significantly compared with the conventional database construction method. Meanwhile, the conventional method needs staying at the landmark point for a while which is time consuming. In our system, we only ask the surveyor to hold the smart-phone walking along the indoor paths once and record the corner reference points, and then using the record data to construct the Wi-Fi database.

To make full use of the recorded corner coordinates, while building the Wi-Fi fingerprints database, we also construct a corner reference trajectory-geomagnetic database (CRTDB) which stores the trajectory segments and the corresponding magnetic signals while passing through corners. It is based on the facts that: 1) the coordinate sequence and the geomagnetic sequence of the trajectory sections passing through the same corner should be similar or consistent with each other; 2) most of the turn corners of official building is right angle. Hence, we can use the CRTDB as the template to match the turning trajectory segments, e.g., testing segments, detected online.

Once the matching is successful, the corner point of the matched template is regarded as the true observation of the testing corner point. Then, we further use this observation to revise the Wi-Fi-PDR fused positioning error.

It is worth noting that CRTDB method is more suitable for corners on the corridor which is simple and unique. In open area, such as hall, the performance of CRTDB method may decrease. Hence, we only establish CRTDB in narrow paths, such as corridor, aisle, stair area, etc. Our experiment proves this limited information can truly improve the whole positioning accuracy. Moreover, our proposed system is mainly used for post-processing; hence it is not strict on processing time overhead. As for the real-time operation, it can be conducted on servers. The smart phone only needs to collect and upload the data, then receive the positioning result from servers.

The experimental designs of most literature are designed by themselves. This lacks fairness and brings difficulties to accuracy comparison. Otherwise, the effectiveness of our system is tested on two open datasets: the off-site track 3 of IPIN2018 competition [19] and the off-site track 3 of IPIN2019 competition [20].

The main contributions of this paper are:

1. We propose a trajectory optimization algorithm which makes use of the landmark observations to optimize the historical positions by means of Gauss-Newton algorithm. It is totally different from the forward-backward smoothing trajectory smoothing algorithm proposed in [21]. Specifically, in our method, the step length and heading of each step of PDR is modeled into the state vector which is optimized by Gauss-Newton method.
2. To our knowledge, we are the first proposing the concept of CRTDB, and we propose a Dynamic Time Warping (DTW) and Pearson correlation coefficient (PCC) combined trajectory matching algorithm to find the pseudo-observation of the corner.
3. Based on the optimized trajectories, the Wi-Fi fingerprint database and CRTDB are constructed. A Kalman Filtering (KF) based method is proposed to fuse the WP, PDR and CRTDB matching results together to obtain an accurate estimation.

The remainder of this paper is organized as follows. Section II introduces the related work and Section III describes the overview of our system. Section IV presents the trajectory optimization algorithm and Wi-Fi database and CRTDB construction. Section V describes the trajectory matching algorithm. Section VI introduces the fusion algorithm of WP, PDR and CRTDB. Section VII describes the experiments and results on the open library. Finally, Section VIII gives conclusion.

II. RELATED WORKS

A. PDR

Benefited from low cost, low power consumption and relatively accurate performance, PDR has become a regular component of most smart-device based positioning systems. The main error of PDR consists in three aspects, namely step detection error, heading error and step length error. Many robust step detection algorithms have been proposed in the past decade, mainly including peak/valley detection [22]–[26],

threshold testing [27] and zero crossing. Heading estimation is mainly based on magnetometer measurements and gyroscope readings [28]. Johan Perul *et al.* [29] leverage the acceleration measurements to help the walking direction estimation via principal component analysis (PCA) and empirical mode decomposition (EDM). The step length is mainly estimated by the beforehand built model such as [23], [25]. A multi-layer perceptron neural network is trained to estimate the step length in [26]. An inverted pendulum model is used to robustly estimate the step length for upper torso mounted IMU [30]. Nevertheless, the false step counting and heading still easily occur for abnormal activities or unfixed devices. And because of the iron magnetic interference and bias of gyroscope, the heading might deviate from the true direction seriously. Hence, mining the deep information of activities and environment plays an indispensable role in PDR systems. For example, Sheng Guo *et al.* [28] trained a decision tree to recognize the motion modes including standing, walking, up/down stairs, taking elevator and door opening. Jin-Shyan Lee *et al.* [27] recognize four smart-phone holding manners, e.g. texting, pocket, swing and calling, to determine the corresponding head updating methods. The recognized motion types help to adjust the parameters in step detection, heading updating and step length calculation. In this paper, the recognized turning corner is used to optimize our PDR trajectories.

B. Wi-Fi Positioning

WP can be divided into two categories according to the basic theory of implementation—ranging based method (RBM) and radio map-based method (RMM). RBM commonly uses the received signal strength (RSS) to estimate the distance between mobile terminal and access point (AP) by constructing a ranging model of RSS [31]. With more than three APs with known locations, RBM is able to determine the positions of terminals in its covered space according to triangulation. In the line-of-sight condition, RBM is able to provide an accurate estimation. But there are enormous obstacles in indoor environment which will cause non-line-of-sight (NLOS) and multi-path problems seriously damaging the performance of RBM. Many works have focused on these two problems and further improved the performance. For example, Yue Yu *et al.* propose a real-time ranging model based on Wi-Fi FTM to reduce the Wi-Fi ranging error caused by clock deviation, non-line-of-sight (NLOS), and multi-path propagation [32]. The results are fine, but it cannot expand to non-FTM supported Aps. With the distance between the user and the AP growing, the error of the ranging model of RSS grows rapidly. Alejandro Correa *et al.* [33] proposed a multi-layer perceptron machine classifier to recognize the distance between user and AP is more or less than 7m, and solving the ranging problem by corresponding means. This may improve the ranging robustness, but it strongly relies on the accuracy of the classifier. The distribution of indoor environment is diverse and complicated, it is hard to eliminate the effect of aforementioned problems of RBM. Besides, it is difficult to know the exact location of all APs. This also limits the application of RBM.

To cope with the aforementioned problems of RBM, RMM is proposed. RMM includes two stages, the training stage and the positioning stage. In training stage, the RSS or the pairwise RSS difference (PSSD) [34] collected from all of the available APs in every known location are stored together to form a fingerprint database. In positioning stage, the RSS vector collected from the around APs is compared to all of the record in the fingerprint database. The most similar records provide the position of the terminal. The commonly used matching approaches are nearest neighbor (NN) algorithm and K-nearest neighbors (KNN) algorithm [35]. Others like hidden Markov model (HMM), deep neural network (DNN) [36], Recurrent Neural Networks (RNN) [37], autoencoder and generative adversarial network (GAN) [38] based algorithms have also been proposed in recent years. These systems directly train deep learning networks to output the positions instead of building the fingerprint database and matching online. RMM has reduced the impact of NLOS and multi-path propagation because it doesn't need ranging. Nevertheless, the training stage and the updating of the database incorporate huge labor-consumption which constrains the extension of RMM. To cope with this defect, crowdsourcing-based Wi-Fi fingerprints construction methods have been proposed by many researchers [39]–[41]. The crowdsourcing-based method can be divided into two categories—active-crowdsourcing method (ACM) and passive-crowdsourcing method (PCM). In ACM, users collect data according to the designed way (e.g. passing the ordered points or paths, keeping a fixed pose while walking...) or equipped with external hardware or software [42] and upload the sampling data to the servers. This process is also labor-consumption. But the acquired data is more reliable. In PCM, data is collected and uploaded automatically by devices without users' participant [25], [43]. This process solves the labor-consumption problem of ACM, but the collected data is relatively larger and unreliable. The SLAM (Simultaneous Localization and Mapping) framework-based systems are kinds of PCM. Hyojeong Shin *et al.* [44] proposed a smart-SLAM system which can automatically construct the Wi-Fi fingerprint database. The fine optimized PDR provides the landmark of the Wi-Fi fingerprint. Liu *et al.* [45] presented a similar solution that incorporates Wi-Fi fingerprint and dead reckoning information for crowdsensing SLAM in unknown indoor environments. Another SLAM framework-based smartphone positioning system is proposed by Qing Liang *et al.* [46]. Its offline phase includes front-end, back-end and map construction. In the front-end stage, the PDR is used to provide the original trajectory, meanwhile the loop-closures are detected by geomagnetical matching and the opportunistic magnetic heading measurement is detected by heading divergence analyzing. The back-end stage leverages the detected information of front-end to construct the optimized trajectory by graph optimization algorithm. With the optimized trajectory and the measured signals from Wi-Fi, geomagnetic and light, the corresponding three fingerprint databases are constructed. Its online positioning stage leverages the constructed database and PDR to provide the real-time positioning. The SLAM based positioning systems can truly reduce the manual labor,

but its accuracy will decrease if the trajectory estimation is not precise enough or the loop detection fails. Liu *et al.* represent fingerprint by a ranked sequence of APs based on the measured RSS without training [47], [48]. The idea is clever and the result is fine, but the positions of APs have to be known before so that the method is actually a little similar to the RBM method.

C. IMU and Wi-Fi Combined Positioning

Pure Wi-Fi based programs each has their own defects because of the intrinsic drawback of WP. Hence, multi-sensors fusion is the realistic solution according to various of works. Wi-Fi and IMU fused methods are proved to be effective [49]. Hybrid data fusion concept and cooperative approach between different localization systems are considered in many works [11]. Yuan Zhuang *et al.* [50] proposed a tightly-coupled algorithm based on EKF to fuse Wi-Fi RSS ranging, INS and PDR together for positioning. And an EKF based two-filter integration of MEMS sensors and Wi-Fi fingerprinting was proposed in their other work [14]. Lyu-Han Chen *et al.* [51] fused the Wi-Fi fingerprint and PDR together using a maximum likelihood-based algorithm. The results showed that the fused method has better positioning accuracy than the PDR system or WP system alone. Among our surveyed papers, particle filtering (PF) and EKF are two most commonly adopted methods to fuse PDR and WP results, such as [25], [44], [46], [52], [54] using PF and [13], [54]–[57] using Kalman filters. Particle filtering is more accurate than EKF on nonlinear optimization problems, but the corresponding computational cost of PF is more expensive than EKF.

The directly fusion of Wi-Fi and PDR results may cause serious errors. For example, the WP result may suddenly jump forward and backward or far away from the last position in a very short time interval. If every WP result were fused with PDR, the final estimated trajectory might deviate seriously from the ground truth and be no longer smooth. Hence, before the fusion of WP results and PDR, the reliability of WP results should be assessed.

D. Geomagnetic Positioning

Geomagnetic is disturbed in indoor environment by things like metal, pillar structures and furniture. The distribution of geomagnetic has a great correlation with the environment. Hence magnetic abnormalities can be used to infer the position. The GM technique has been intensively researched for positioning. The geomagnetic profile sampled online is matched with the GF database collected offline to find the position. The generally used matching method is Dynamic Time Warping (DTW) [58]. Qiu *et al.* combine DTW and Spearman's correlation coefficient to improve the matching performance [59]. There may be different places that have similar magnetic features in indoors. This may result in user's position suddenly jumping to another place. To alleviate this issue, GM positioning system commonly employs filtering methods, such as particle filter [60], or integrates with other techniques, such as PDR and wireless localization [15] to be realized. However, modeling of the GM is very challenging. The traditional process of collecting geomagnetic fingerprint

information needs expensive and specialized instruments. It is time-consuming and labor-intensive [61]. To reduce the burden of GF database establishment, the walk-survey method [15], [46] and crowdsourcing strategies [62] is adopted by some researches. In this paper, we also aim to reduce the burden of fingerprints database establishment.

III. SYSTEM OVERVIEW

In our system, there are two databases needed to be built, e.g., corner reference trajectory database (CRTDB) and the Wi-Fi fingerprint database. In the training stage, the surveyor is holding a smart-phone walking on the passage in the indoor environment. Meanwhile, the corner points can be recorded manually once the surveyor passing through and at least two ground truth points need to be recorded for each training path. This work can be conducted in an active crowdsourcing manner (ACM). For example, we let the participants walk along the designed paths sequentially passing through specific corners and make records on the corners via just click the smart-phone screen. This process is easily implemented. You can refer to the data collection app used by IPIN: GetSensorData20 (https://github.com/lopsi/GetSensorData_Android)

After getting the training data, the PDR is implemented to generate the original trajectories. Then, our proposed trajectory optimization algorithm is used to optimize the trajectories via fusing the recorded corner landmarks. The positions of optimized trajectories are regarded as the labels of the Wi-Fi radio signals sampled in the corresponding time. Then, the labels and its corresponding Wi-Fi RSS are together stored into the fingerprint database. In this way, the Wi-Fi fingerprint databases of the buildings are constructed.

As for the building of CRTDB, the corner points need to be extracted firstly from the optimized trajectories for IPIN datasets. The turning action detection method is described in Section VI-B. Then, the trajectory segment containing 10 steps around each turning point and the corresponding geomagnetic modulus are combined to form a record stored in the CRTDB. When collecting training data by yourself, you can plan the path in advance. And the data collection APP can be designed to manually record (for example, tap the mobile phone screen) the timestamp passing the corner point, so that the precise timestamp of the corner point in each trajectory can be known.

In the testing stage, PDR and Wi-Fi fingerprint positioning are fused by KF for real-time location. Once the turning event is detected, the dynamic time warping (DTW) and Pearson correlation coefficient (PCC) combined trajectory matching algorithm is proposed to find the true observation of the corner. Then, the estimated trajectory via PDR and Wi-Fi is further corrected by the found reference point of turning corner. Furthermore, the optimized trajectories on testing stage can be used to update the Wi-Fi fingerprint database if possible. Fig. 1 shows the architecture of our proposed system.

IV. TRAJECTORY OPTIMIZATION AND DATABASE CONSTRUCTION

In this section, we will introduce our walking trace generation method. Firstly, horizontal PDR is optimized by the recorded reference points. The goal is to correct the horizontal

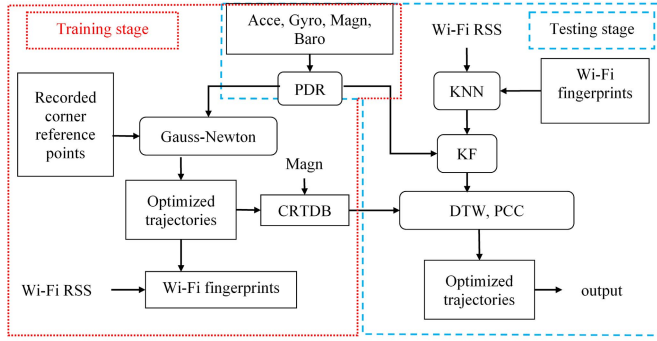


Fig. 1. The block diagram of the proposed positioning system. For detailed description of the algorithm flow, please refer to section III. In training stage, the original PDR is optimized by the reference points using Gauss-Newton algorithm. Then the Optimized PDR trajectories provide the position label for Wi-Fi fingerprint database and CRTDB. In testing stage, PDR is fused with Wi-Fi fingerprint positioning in KF. Then the DTW and PCC combined algorithm is used to find the corner points which is used to further optimize the PDR and Wi-Fi fused results.

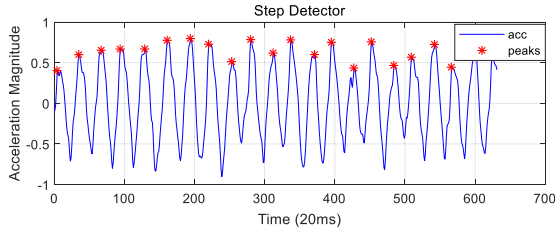


Fig. 2. Step detection results.

PDR errors and generate a relatively accurate trajectory for Wi-Fi database construction. Then, the layer estimation of PDR trajectory based on barometer is mentioned simply.

A. Original 2-D PDR

Smart-phone based PDR mainly consists of three components e.g., step detection, step length estimation and heading determination. For the step detection, the acceleration magnitude is smoothed by low pass filtering. Then we utilize a peak detection-based detector, commonly used in many studies, to count the walking steps and Fig. 2 illustrates the step detection result. Afterwards, the step frequency is used to estimate the step length. The step length model proposed in [63] is adopted. As for the step heading, it is from the attitude solution estimated by fusing angular rate and magnetometer output. Then the 2-D PDR trajectory can be built as follows:

$$P_k = \begin{bmatrix} x_k \\ y_k \end{bmatrix} = \begin{bmatrix} x_{k-1} \\ y_{k-1} \end{bmatrix} + sl_k * \begin{bmatrix} \sin \theta_k \\ \cos \theta_k \end{bmatrix}. \quad (1)$$

in which, P_k is the coordinate of k -th step. sl_k and θ_k are the step length and step heading of the k -th step respectively.

B. 2-D PDR Trajectory Model

The 2-D PDR trajectory can be divided into several segments by reference points recorded on the trace. The end point coordinate of every segment is estimated as follows:

$$P_{seg} = P_0 + \sum_{i=1}^N sl_i * \begin{bmatrix} \sin \theta_i \\ \cos \theta_i \end{bmatrix}. \quad (2)$$

in which, P_{seg} is the end-point position estimated by PDR. N is the step number of the trace segment. The error of P_{seg} is resulted by the step length error and the heading error of every step. Hence, we construct the state vector using every step length and step heading, that is:

$$X = [sl_1, sl_2, sl_3, \dots, sl_N, \theta_1, \theta_2, \theta_3, \dots, \theta_N], \quad \text{where } sl_i \geq 0, -\pi \leq \theta_j \leq \pi \quad (3)$$

The state equation is same as equation (2), and the objective equation is as follows:

$$\min_X f(X) = \frac{1}{2} * (P_{seg} - Z_m)^2. \quad (4)$$

where, Z_m is the true coordinate of the reference point. The residual error term can be expressed as:

$$\varepsilon = P_{seg} - Z_m = P_0 + \sum_{i=1}^N sl_i * \begin{bmatrix} \sin \theta_i \\ \cos \theta_i \end{bmatrix} - Z_m. \quad (5)$$

The equation (4) is a nonlinear function about the state vector X , and needed to be linearized. The Jacobi matrix of equation (5) respect to X can be represented as:

$$J(X) = \begin{bmatrix} \sin \theta_1, \dots, \sin \theta_N, sl_1 * \cos \theta_1, \dots, sl_N * \cos \theta_N \\ \cos \theta_1, \dots, \cos \theta_N, sl_1 * -\sin \theta_1, \dots, sl_N * -\sin \theta_N \end{bmatrix}. \quad (6)$$

The Second-order Taylor expansion of the objective equation is as follows:

$$f(X) \approx f(X_0) + \left(J(X_0)^T * \varepsilon_0 \right)^T * \delta_X + \frac{1}{2} \delta_X^T * \left(J(X_0)^T * J(X_0) + S \right) * \delta_X. \quad (7)$$

where, X_0 is function expansion point. T means the transposing operation. S is the second-order information term which is omitted in Gauss-Newton algorithm. $\delta_X = X - X_0$ is the distance of X relative to the expansion point X_0 . ε_0 is the residual at X_0 .

Gauss-Newton algorithm is an iterative solution process. The k^{th} iteration is as follows:

$$X_{k+1} = X_k - \left(J(X_k)^T * J(X_k) \right)^{-1} J(X_k)^T * \varepsilon_k. \quad (8)$$

With the model above, the Gauss-Newton algorithm can be used to fuse the reference-point observation to optimize the PDR estimation. Afterwards, the trajectory is reconstructed with the optimized state vector X using equation (1). The corresponding algorithm is showed in Algorithm 1. It is deserved to mention that though Gauss-Newton is used in this paper as the optimization method, other optimization algorithm, such as Levenberg-Marquardt algorithm, can also be used to solve this optimization problem and handle the singularity problem of Gauss-Newton algorithm. We present a Gauss-Newton based solution in Algorithm 1. And Fig. 3 illustrates the effectiveness of algorithm 1. We can see that our proposed trajectory optimization algorithm can effectively correct the whole trajectory by properly adjust the step headings and step lengths.

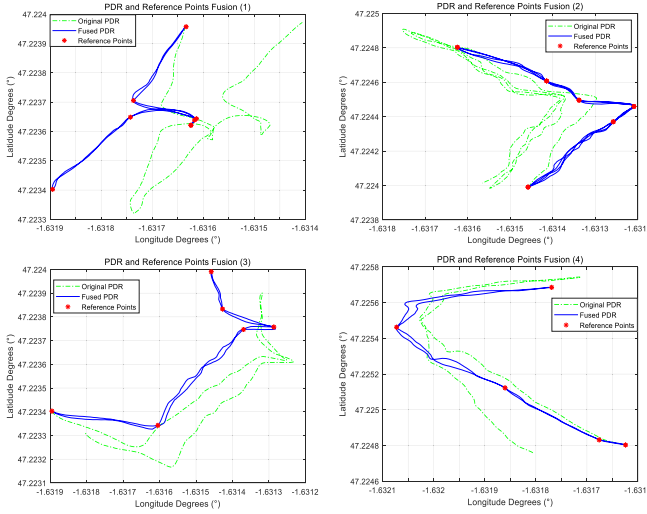


Fig. 3. The effect of PDR and reference points fusion.

Algorithm 1 Trajectory Optimization Based on Gauss-Newton

Input: Initial position: P_0 ; Step lengths and headings from PDR: $[sl_1, sl_2, \dots, sl_n, \theta_1, \theta_2, \dots, \theta_n]^T$; Corner / Reference Point: Z_m

Output: Optimized Trajectory: P_{opt}

```

1: Initialize:  $X \leftarrow [sl_1, sl_2, \dots, sl_n, \theta_1, \theta_2, \dots, \theta_n]^T$ ;
    $K = 10$ ; // Parameter initialization
2: for  $k = 1: K$  do
3:    $\epsilon = P_{seg}(X) - Z_m = \left( P_0 + \sum_{i=1}^N sl_i * \begin{bmatrix} \sin \theta_i \\ \cos \theta_i \end{bmatrix} \right) - Z_m$ ; // Computing the residual at k-th step
4:    $J = \begin{bmatrix} \sin \theta_1 & \sin \theta_2 & \dots & \sin \theta_n & sl_1 * \cos \theta_1 & sl_2 * \cos \theta_2 & \dots & sl_n * \cos \theta_n \\ \cos \theta_1 & \cos \theta_2 & \dots & \cos \theta_n & sl_1 * -\sin \theta_1 & sl_2 * -\sin \theta_2 & \dots & sl_n * -\sin \theta_n \end{bmatrix}$ ;
5:    $X = X - (J^T J)^{-1} J^T \epsilon$ ; // State update
6:    $[sl_1, sl_2, \dots, sl_n, \theta_1, \theta_2, \dots, \theta_n]^T \leftarrow X$ ;
7:   if  $\|\epsilon\| < \sigma$  then
8:     break;
9:   end if
10: end for
11:  $P_{opt}(0) = P_0$ ;
12: for  $i = 1: n$  do
13:    $P_{opt}(i) = P_{opt}(i-1) + sl_i * \begin{bmatrix} \sin \theta_i \\ \cos \theta_i \end{bmatrix}$ ;
14: end for

```

C. Layer Estimation by Barometer

Barometer reading is commonly used to estimate the height or layer of PDR. We used the height estimation method described in a given paper [64] in which the relationship between height and atmospheric pressure can be represented by following:

$$H = a * \left(\frac{P}{c} \right)^b - H_0. \quad (9)$$

where H is height relative to the initial ground, P is atmospheric pressure measurement, H_0 is the height of initial ground.

$$H_0 = a * \left(\frac{P_0}{c} \right)^b. \quad (10)$$

where a , b and c are constant parameters, where a is set as 44330, b is 0.19026 and c is 101325 according to [64].

To smooth the height estimation result, we leverage Kalman Filtering (KF) to filter the original height. The state equation is as:

$$\begin{bmatrix} h_k \\ hr_k \end{bmatrix} = \begin{bmatrix} 1 & \delta t \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} h_{k-1} \\ hr_{k-1} \end{bmatrix} + q_k. \quad (11)$$

where h_k is the height of time k , hr_k is the height changing rate of time k , and δt is the time interval. q_k is Gaussian white noise. The measurement equation is as:

$$hz_k = h_k + r_k. \quad (12)$$

where hz_k is calculated by equation (8) using the barometer measurement of time k . r_k is Gaussian white noise.

The estimated height is then used to determine the layer of the trajectory. And the layer is estimated as:

$$L_k = \text{round} \left(\frac{h_k}{\Delta} \right). \quad (13)$$

The function of $\text{round}(\cdot)$ means getting the integer part of the input parameter. Δ is the height per layer determined in training stage. With the layer estimation, the 2D-PDR can be upgraded to 3D-PDR.

V. FINDING CORNER REFERENCE POINT BY TRAJECTORY MATCHING

This section provides our proposed trajectory segment matching algorithm based on the built CRTDB. Firstly, we introduce the DTW and PCC. Then the details of our algorithm are described.

Dynamic Time Warping: There are many similarity or distance functions in time series data, the most prominent of which is DTW. The algorithm is based on the idea of dynamic programming (DP), which solves the problem of template matching with different lengths. It is a relatively early and classic algorithm in speech recognition. In the field of indoor positioning, DTW is commonly used in geomagnetical positioning techniques, such as [58]. The common features between speech sequence and geomagnetic sequence are: they are both time series data, and even the data length of template is usually different from the data length of matching sequence, it is able to provide a successful matching result. In this paper, the trajectory segments are kind of time series data. Every sequence of template trajectory and trajectory to be matched has the same length. Hence, it is feasible to use DTW for our trajectory matching.

Pearson Correlation Coefficient: it is used to evaluate the linear correlation of two statistical variables. It reflects the direction and degree of the change trend between two variables, and its value ranges from -1 to $+1$. 0 means that the two variables are not linear correlated, and positive values mean positive correlation, and negative values mean negative correlation. The larger the absolute value, the stronger the correlation. In the situation of our trajectory matching, if the two trajectories which path through two similar routes centered by turning corners in a same direction, their PCC would be highly approaching $+1$.

The combination of DTW and Spearman's correlation coefficient has been proved more effective in geomagnetic matching positioning in [59]. DTW is sensitive to distance between two variables, while PCC is sensitive to correlation. Thus, DTW is more likely trying to find the nearest sequence but not relatively care about correlation. PCC is trying to find the most correlative sequence but not care about the distance. In this paper, we are trying to find a both near and correlative template sequence in every matching stage. Hence, the combination of DTW and PCC is more appropriate to solve our problem.

A. Trajectory Matching Based on DTW

The trajectories are three-dimensional sequences. The template sequence is defined as:

$$S_{temp} = \begin{bmatrix} x_1 & x_{k-m+1} & \cdots & x_k & x_{k+1} & \cdots & x_{k+m} \\ y_1 & y_{k-m+1} & \cdots & y_k & y_{k+1} & \cdots & y_{k+m} \\ z_1 & z_{k-m+1} & \cdots & z_k & z_{k+1} & \cdots & z_{k+m} \end{bmatrix}. \quad (14)$$

The testing sequence is represented as:

$$S_{test} = \begin{bmatrix} \chi_1 & \chi_{k-m+1} & \cdots & \chi_k & \chi_{k+1} & \cdots & \chi_{k+m} \\ \Upsilon_1 & \Upsilon_{k-m+1} & \cdots & \Upsilon_k & \Upsilon_{k+1} & \cdots & \Upsilon_{k+m} \\ \mathcal{Z}_1 & \mathcal{Z}_{k-m+1} & \cdots & \mathcal{Z}_k & \mathcal{Z}_{k+1} & \cdots & \mathcal{Z}_{k+m} \end{bmatrix}. \quad (15)$$

where, the $[x_k, y_k]^T$ is the coordinate of k-th point. z_k is the geomagnetic norm value on the k-th point. Then the distance matrix of coordinate sequence d^{co} of S_{temp} and S_{test} is calculated as:

$$d_{ij}^{co} = (\chi_i - x_j)^2 + (\Upsilon_i - y_j)^2, \quad \text{where } 1 \leq i, j \leq 2m+1. \quad (16)$$

in which d_{ij}^{co} is i^{th} row and j^{th} column element of d^{co} . The distance matrix of geomagnetic sequence d^{ge} of S_{temp} and S_{test} is calculated as:

$$d_{ij}^{ge} = (\mathcal{Z}_i - z_j)^2, \quad \text{where } 1 \leq i, j \leq 2m+1. \quad (17)$$

Then the cumulative matrix D^{co} of d^{co} is calculated as:

$$D_{ij}^{co} = \begin{cases} d_{ij}^{co}, & i = j = 1 \\ d_{ij}^{co} + D_{i-1,j}^{co}, & 2 \leq i \leq 2m+1 \\ & \text{and } j = 1 \\ d_{ij}^{co} + D_{i,j-1}^{co}, & i = 1 \text{ and } 2 \leq j \leq 2m+1 \\ d_{ij}^{co} + \min \{D_{i-1,j}^{co}, D_{i,j-1}^{co}, D_{i-1,j-1}^{co}\}, & 2 \leq i, j \leq 2m+1. \end{cases} \quad (18)$$

And the cumulative matrix D^{ge} of d^{ge} is calculated as:

$$D_{ij}^{ge} = \begin{cases} d_{ij}^{ge}, & i = j = 1 \\ d_{ij}^{ge} + D_{i-1,j}^{ge}, & 2 \leq i \leq 2m+1 \\ & \text{and } j = 1 \\ d_{ij}^{ge} + D_{i,j-1}^{ge}, & i = 1 \text{ and } 2 \leq j \leq 2m+1 \\ d_{ij}^{ge} + \min \{D_{i-1,j}^{ge}, D_{i,j-1}^{ge}, D_{i-1,j-1}^{ge}\}, & 2 \leq i, j \leq 2m+1. \end{cases} \quad (19)$$

The output of DTW is the final elements of D^{co} and D^{ge} : $dd^{co} = D_{2m+1,2m+1}^{co}$ and $dd^{ge} = D_{2m+1,2m+1}^{ge}$. The smaller the output of DTW, the more matching the corresponding matching template and the sequence to be matched in the view of DTW. However, as mentioned before, DTW is trying to matching the nearest trajectory but not relatively care about correlation. The smallest output of DTW may not be the true what we wanted. Hence, we leverage DTW to select several candidate sequences, then use PCC to choose the best matching one. We use a threshold-based method to select the candidates whose DTW results are lower than the set threshold.

$$d_{flag} = \begin{cases} 1, & dd^{co} < \sigma_d^{co} \text{ and } dd^{ge} < \sigma_d^{ge} \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

where σ_d^{co} and σ_d^{ge} are the set thresholds. Only the sequence satisfying the above condition can be regarded as candidate and participate the next judgement process. Meanwhile the sequence with lower DTW result is regarded as better candidate.

B. Trajectory Matching Based on PCC

The candidates filtered by DTW then are assessed again by PCC. The PCC of the two matching sequences is calculated as follows:

$$\begin{aligned} \text{corrcoef}(V_{temp}, V_{test}) &= \frac{\sum_{i=1}^M (V_{temp}^i - \overline{V_{temp}}) * (V_{test}^i - \overline{V_{test}})}{\sqrt{\sum_{i=1}^M (V_{temp}^i - \overline{V_{temp}})^2} * \sqrt{\sum_{i=1}^{2m+1} (V_{test}^i - \overline{V_{test}})^2}}. \end{aligned} \quad (21)$$

in which, i refers to i^{th} element of the vector V_{temp} or V_{test} . Here we calculate the horizontal and vertical correlation of two matching sequences separately. Then the horizontal, vertical and geomagnetic correlation r_h , r_v , r_g are as follows:

$$r_h = \text{corrcoef}(V_{x,temp}, V_{x,test}). \quad (22)$$

$$r_v = \text{corrcoef}(V_{y,temp}, V_{y,test}). \quad (23)$$

$$r_g = \text{corrcoef}(V_{z,temp}, V_{z,test}). \quad (24)$$

And the production of r_h , and r_v is:

$$\rho_{hv} = r_h * r_v. \quad (25)$$

The production of r_h , and r_g is:

$$\rho_{hg} = r_h * r_g. \quad (26)$$

$V_{x,temp}$ and $V_{\chi,temp}$ are the horizontal components of S_{temp} and S_{test} respectively, and $V_{y,temp}$ and $V_{\gamma,temp}$ are the vertical components, and $V_{z,temp}$ and $V_{Z,temp}$ are the geomagnetic components. If the horizontal and vertical components are positively or negatively correlated at the same time, their production ρ_{hv} is positive, otherwise it is negative. The positive result means that the two sequences have a similar shape, and the closer the value is to 1, the more similar the shape. Meanwhile the geomagnetic correlation should be coincided with the coordinate correlation for two matched sequences, thus ρ_{hv} and ρ_{hg} should be positive at the same time. And the two matched sequences should significantly correlate with each other means that the r_h , r_v and r_g are close to 1 or -1 at the same time. Hence only the sequences satisfy the following conditions are considered for further judgement:

$$COR_{flag} = \begin{cases} 1, & r_h > \alpha \text{ and } r_v > \beta \text{ and } r_g > \gamma \text{ and} \\ & \rho_{hv} > 0 \text{ and } \rho_{hg} > 0 \\ 0, & \text{otherwise.} \end{cases} \quad (27)$$

where α , β and γ are the set thresholds. First, we use DTW to find the sequences that satisfy the set threshold. Then PCC is used to further filter out candidates with relatively small correlation. Finally, the we propose the following judgement condition composed by DTW and PCC results for choosing the final matching result S_{out} :

$$S_{out} = \underset{S_{temp}}{\operatorname{argmin}} \frac{1}{\rho_{hv} * \rho_{hg}} * \frac{dd^{co}}{\mu^{co}} * \frac{dd^{ge}}{\mu^{ge}}. \quad (28)$$

in which, μ^{co} and μ^{ge} are parameters used to determine the weight of DTW and PCC. The larger the $\mu^{co} * \mu^{ge}$, the greater the weight of DTW, and vice versa, the greater the weight of PCC. In this equation, we are trying to find the sequence with relatively larger ρ_{hv} , ρ_{hg} and smaller dd^{co} and dd^{ge} .

The algorithm is shown in Algorithm 2. The functions, DTW and *corrcoef*, shown in the Algorithm 2 are described separately in part A and B above, and the function *middle* gets the middle element of its input sequence. Fig. 4 shows the effect of Algorithm 2.

VI. POSITIONING BASED ON PDR, WI-FI, AND CRTDB

In this section, we will introduce our positioning system which fuses PDR and Wi-Fi database for real-time positioning and is revised by corner reference-points matched by using the Algorithm 2 introduced in section V.

A. PDR and Wi-Fi Fused Positioning

We leverage KNN based means for WP. But its results may not reliable for reasons such as NLOS, signal fluctuation of devices, dynamic environmental occlusion, insufficient samples of database etc. Therefore, it is necessary to evaluate the reliability of WP results in advance in order to decide whether to use it for fusion with PDR. The following three conditions are used to reject the WP results.

Algorithm 2 Trajectory Matching Based on DTW and PCC

Input: The database: CRTDB; The testing sequence: S_{test}
Output: The found corner's coordinate: P_{corner}
1: **Initialize:** $S_{out} = []$; $\Delta = initmax$; // Initialize temp variables
2: **for each** S_{temp} **in** CRTDB **do**
3: $[dd^{co}, dd^{ge}] = DTW(S_{temp}, S_{test})$;
4: **if** $dd^{co} < \sigma^{co}$ **and** $dd^{ge} < \sigma^{ge}$ **then**
5: $r_h = corrcoef(V_{x,temp}, V_{\chi,temp})$;
6: $r_v = corrcoef(V_{y,temp}, V_{\gamma,temp})$;
7: $r_g = corrcoef(V_{z,temp}, V_{Z,temp})$;
8: $\rho_{hv} = r_h * r_v$;
9: $\rho_{hg} = r_h * r_g$;
10: **if** $r_h > \alpha$ **and** $r_v > \beta$ **and** $r_g > \gamma$ **and** $\rho_{hv} > 0$ **and** $\rho_{hg} > 0$ **then**
11: $r = \frac{1}{\rho_{hv} * \rho_{hg}} * \frac{dd^{co}}{\mu^{co}} * \frac{dd^{ge}}{\mu^{ge}}$;
12: **if** $r < \Delta$ **then**
13: $\Delta = r$;
14: $S_{out} = S_{temp}$;
15: **end if**
16: **end if**
17: **end for**
18: **end for**
19: $P_{corner} = middle(S_{out})$;

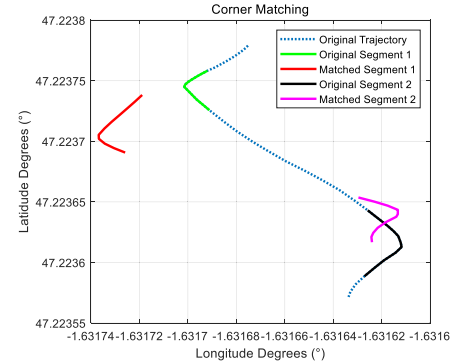


Fig. 4. An example of corner matching effect. The corner points of the matched segments are regarded as the observation of the corresponding corner points of original segments. Then the original trajectory is revised by those observations.

Firstly, the layer of WP result is not consistent with the one determined by barometric pressure. Secondly, the angle between the two vectors, V_w and V_p , which are from the last fusion point to current WP point and PDR positioning point separately, should exceed a certain threshold. Thirdly, the distance, d_{wp} , between WP point and PDR positioning point should exceed a certain dynamic threshold. The combined judgement condition is expressed as following:

$$WP_{flag} = \{L_w \neq L_p \text{ or } \Theta(V_w, V_p) > \vartheta \text{ or } d_{wp} > \tau\}. \quad (29)$$

in which, L_w is the layer of WP point and L_p is the layer determined by barometric pressure. $\Theta(V_w, V_p)$ means calculating the angle between V_w and V_p . ϑ and τ are the set thresholds. τ is proportional to the step number counted since

last fused point.

$$\tau = a * Step_{num}. \quad (30)$$

in which, a is a set constant.

The admitted WP result is then fused with PDR in KF. The WP result is regarded as observation and the PDR trajectory segment from last fused point to current point estimated by PDR is seen as the estimation.

At k -th time point, the system state is the 2-D coordinate of PDR:

$$\zeta_k = [p_k^x, p_k^y]^T. \quad (31)$$

The state transition equation is:

$$\zeta_{k+1} = A * \zeta_k + sl_k * [\sin \theta_k, \cos \theta_k]^T + w_k. \quad (32)$$

where, sl_k is the k -th step length and θ_k is the corresponding heading. w_k is the system noise, a Gaussian white noise. A is 2×2 identity matrix.

And the measurement equation is:

$$\zeta_k^{wifl} = B * \zeta_k + v_k. \quad (33)$$

where, ζ_k^{wifl} is the position provided by Wi-Fi matching. v_k is the measurement Gaussian white noise. B is 2×2 identity matrix.

The state prediction in KF is:

$$\bar{\zeta}_{k+1} = A * \zeta_k + sl_k * [\sin \theta_k, \cos \theta_k]^T. \quad (34)$$

$$\bar{P}_{k+1} = A * P_k * A^T + Q. \quad (35)$$

where, P_k is the covariance of the estimation, indicating the uncertainty of the estimation state. Q is the covariance of the system process.

If the system has not received the result of Wi-Fi positioning, only the state prediction is performed. Once the Wi-Fi positioning result is received, the state updating process is:

$$K_{k+1} = \bar{P}_{k+1} * B^T * (B * \bar{P}_{k+1} * B^T + R)^{-1}. \quad (36)$$

$$\zeta_{k+1} = \bar{\zeta}_{k+1} + K_{k+1} * (\zeta_{k+1}^{wifl} - B * \bar{\zeta}_{k+1}). \quad (37)$$

$$P_{k+1} = (I - K_{k+1} * B) * \bar{P}_{k+1}. \quad (38)$$

where, K_{k+1} is the Kalman gain. R is the covariance of measurement noise.

B. Turning Corner Detection

In testing stage, the consecutive m steps around a corner, which are fused by PDR and Wi-Fi, are leveraged to form a sequence, S_{latest} , for CRTDB matching. First of all, the turning action should be detected. In this paper, the heading changing is used to judge whether a turning action is occurred. The following condition is used for this task:

$$|Head_{m-\kappa} - Head_{\kappa}| > \eta. \quad (39)$$

in which, $Head_{\kappa}$ means the heading of κ^{th} step in S_{latest} and $1 < \kappa < m/2$. m is the length of S_{latest} . η is a set threshold. The middle step of S_{latest} is regarded as where the turning action happens. If the condition is met, it means S_{latest} may be a trajectory centered by a turning corner. Then,

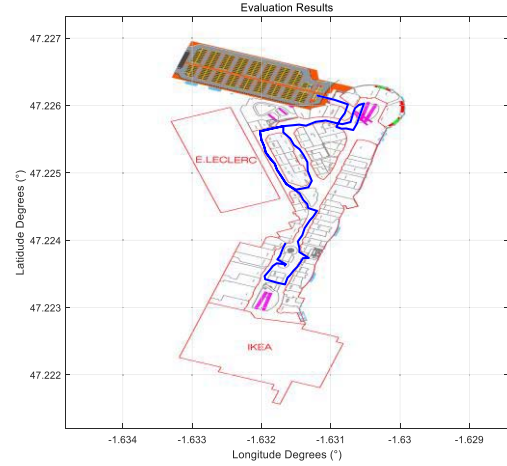


Fig. 5. The ground truth of the evaluation set of IPIN 2018.

Algorithm 2 is used to find the pseudo corner reference-point observation. If S_{latest} is matched successfully by Algorithm 2, then the historical trajectory from the last found point to current matched point is optimized by Algorithm 1. Otherwise, the corresponding turning action is not considered to occur at a corner.

The final results fused by PDR, Wi-Fi and corner reference-points are then composed with the Wi-Fi RSS to update the Wi-Fi fingerprint database.

VII. EXPERIMENTAL VALIDATION

We evaluate our proposed algorithms on an open dataset, e.g., the off-site track 3 of IPIN2018 and IPIN2019 competitions [65], which includes training set, validation set and evaluation set with different types of data. The phone used for data sampling is Samsung SM-A520F, and API Android version is 24. The whole data processing is conducted on a PC with a CPU of Intel(R) Xeon(R) E-2224G and a 16.0GB RAM. We use Matlab2020b for programing. The training set and the validation set contain the landmark points while evaluation set not. We leverage Wi-Fi RSS data, geomagnetic data, IMU data and landmarks provided in training set and validation set to build our Wi-Fi fingerprint database and CRTDB. The performance of our system is assessed by the evaluation set. For instance, Fig. 5 shows the ground truth trajectory of evaluation set of IPIN2018. In order to evaluate the performance, IPIN officially published the true value and the corresponding timestamp of each evaluation point of each evaluation set, as well as the tools used to assess the results [19], [20]. Our experimental results are obtained by using official real values and the evaluation tools.

Firstly, we use Algorithm 1 to process the IMU data and the landmarks provided in training and validation sets to generate reference trajectories. Then we build the Wi-Fi fingerprint database and CRTDB. Finally, we assess our system on the evaluation sets. Fig. 6 shows the only Wi-Fi fingerprint-based positioning results and the original PDR estimation on the evaluation set of IPIN2018. Fig. 7 shows the only Wi-Fi fingerprint-based results and the original PDR estimation on the evaluation set of IPIN2019. We can see that

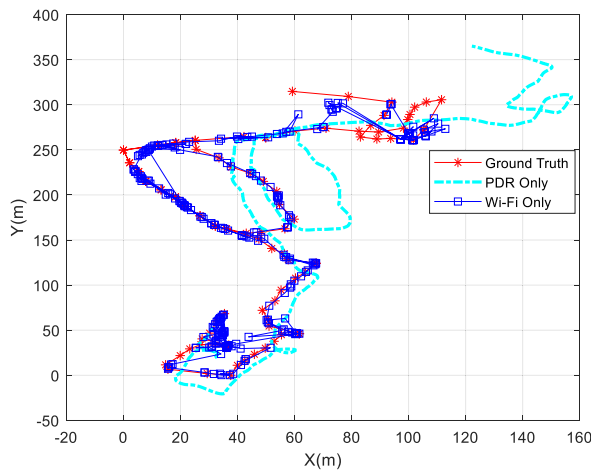


Fig. 6. Original Wi-Fi and PDR Results of IPIN 2018. To be more intuitive, we translate the geodetic coordinate in degree into local coordinate in meter.

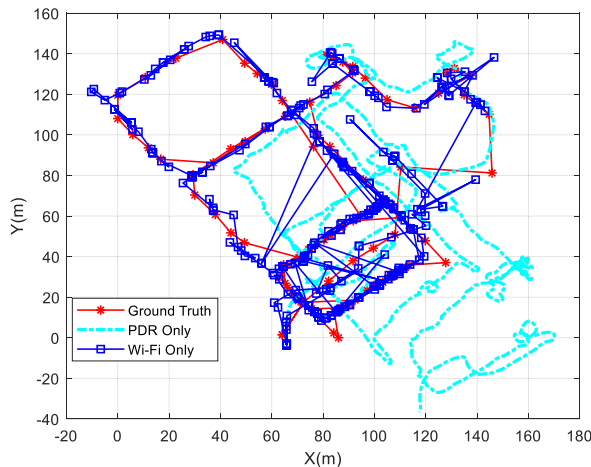


Fig. 7. Original Wi-Fi and PDR Results of IPIN 2019. To be more intuitive, we translate the geodetic coordinate in degree into local coordinate in meter.

Wi-Fi positioning results don't drift over time like PDR. But PDR trajectory is smoothing and consecutive while Wi-Fi positioning results randomly jump around the true position. The complementary characteristics of Wi-Fi positioning and PDR make the result of the fusion better than any result alone. Besides, we can also know from Fig. 6 that Wi-Fi positioning is more accurate in corridor than in open areas. This is because the Wi-Fi RSS data in corridor are more different from each other than in open areas. The only Wi-Fi fingerprint-based positioning results on IPIN2019 are not stable like that on IPIN2018. This indicates that the accuracy of only Wi-Fi fingerprint-based positioning varies greatly in different environments.

Table I and Table II show the final results of our proposed indoor positioning system. We can find that the proposed Wi-Fi and PDR fused system achieves an accuracy with mean error of 5.88m and 6.76m respectively for IPIN2018 and 2019 data sets. Meanwhile, our proposed Wi-Fi, PDR and CRTDB fused system realizes an accuracy with mean error of 4.93m and 5.4m. With CRTDB optimizing, the median errors are

TABLE I
EXPERIMENTAL ERROR RESULTS OF IPIN2018

Algorithm	Median (m)	Mean (m)	RMS (m)	90th perc(m)	75th perc(m)
PDR, Wi-Fi Fusion	5.04	5.88	7.28	12.30	7.05
PDR, Wi-Fi, CRTDB Fusion	3.34	4.93	5.88	8.45	5.84

TABLE II
EXPERIMENTAL ERROR RESULTS OF IPIN2019

Algorithm	Median (m)	Mean (m)	RMS (m)	90th perc(m)	75th perc(m)
PDR, Wi-Fi Fusion	6.17	6.76	7.86	10.82	8.41
PDR, Wi-Fi, CRTDB Fusion	4.5	5.4	6.79	10.78	7.01

improved by more than 33% and 27% respectively compared to that without CRTDB. The error cumulative distribution function (CDF), showing in Fig. 8, reveals that our proposed system has an accuracy with 75% of errors less than 5.84m and 7.01m respectively. Compared with the original PDR and pure Wi-Fi positioning results, the fused positioning results are obviously improved on both datasets. The comparison of the results before and after the using of CRTDB proves that CRTDB can indeed improve the positioning accuracy. The error distance comparison at each evaluation point shows in Fig. 9. It reveals that PDR and Wi-Fi fusion results are better than that of pure PDR or pure Wi-Fi, and the accuracy of PDR, Wi-Fi and CRTDB fusion is more accurate than that of PDR and Wi-Fi fusion. The comparison of the results before and after the fusion shows that our proposed PDR, Wi-Fi and CRTDB fusion algorithm is effective.

Our experimental results achieve a similar accuracy to the conventional Wi-Fi fingerprint-based positioning accuracy which needs a labor-intensive database construction process. This proves that our proposed Algorithm 1 is effective and its optimized trajectory can indeed be used to build the Wi-Fi database and CRTDB. Fig. 10 and Fig. 11 illustrate the trajectory estimation results of our proposed system. We can see that the CRTDB matching corrects some corner points to the true corner points, e.g., *Corner 1*, *Corner 5* and *Corner 6* in Fig. 10. This also means that our proposed Algorithm 2 is effective. The layer estimation results of IPIN2018 and IPIN2019 are showed in Fig. 12. It illustrates that the only Wi-Fi estimated layer fails in some cases, such as the evaluation points 81 to 89 of IPIN2018 and the 43rd evaluation point of IPIN2019. However, the floor error estimated by Wi-Fi and barometer fusion is zero on both IPIN2018 and IPIN2019. This can help our algorithm to reject some the wrong Wi-Fi positioning input in PDR and Wi-Fi fusion stage. And also, it implements a 3D positioning. We can see that our PDR and Wi-Fi fused result is smooth

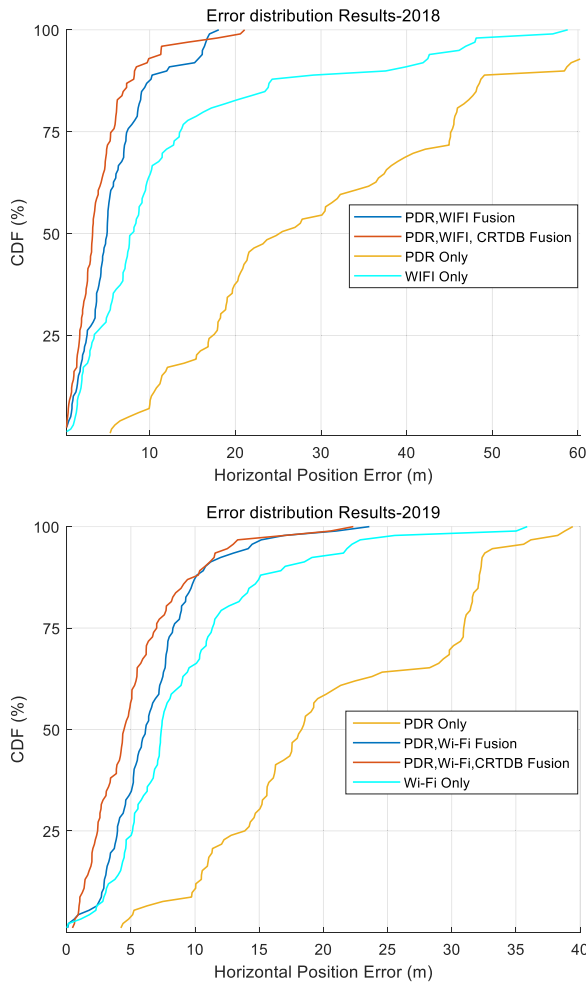


Fig. 8. The comparison of CDF errors distribution on two datasets.

and accurate. This can be own to our proposed Wi-Fi and PDR fusion algorithm and Wi-Fi screening method.

However, there are some mismatched corner points, such as *Corner 2*, *Corner 3* and *Corner 4* in Fig. 10. This is mainly caused by two reasons. On the one hand, if the testing trajectory dose not pass through the same corner point in a same way with the recorded trajectories in CRTDB, it will result in matching failed, such as *Corner 3*, or forcibly pulling the corner point to the recorded corner point such as *Corner 2*. On the other hand, if there are several similar corners in a small area and the geomagnetic distributions are also similar, it may result in wrong matching, such as *Corner 4*. The first situation can be solved by recording more possible turning ways. The second situation is more difficult to be solved, and it may need other information to distinguish the different corners. In our future work, we will try to solve these problems to realize a more robust and accurate matching result. By the way, our proposed CRTDB matching method is not suitable for open area or outdoor situations. Because in these environments the turning freedom is too large to result in large wrong matching.

Although, our proposed CRTDB matching concept has several drawbacks, our experiment shows that in most of the

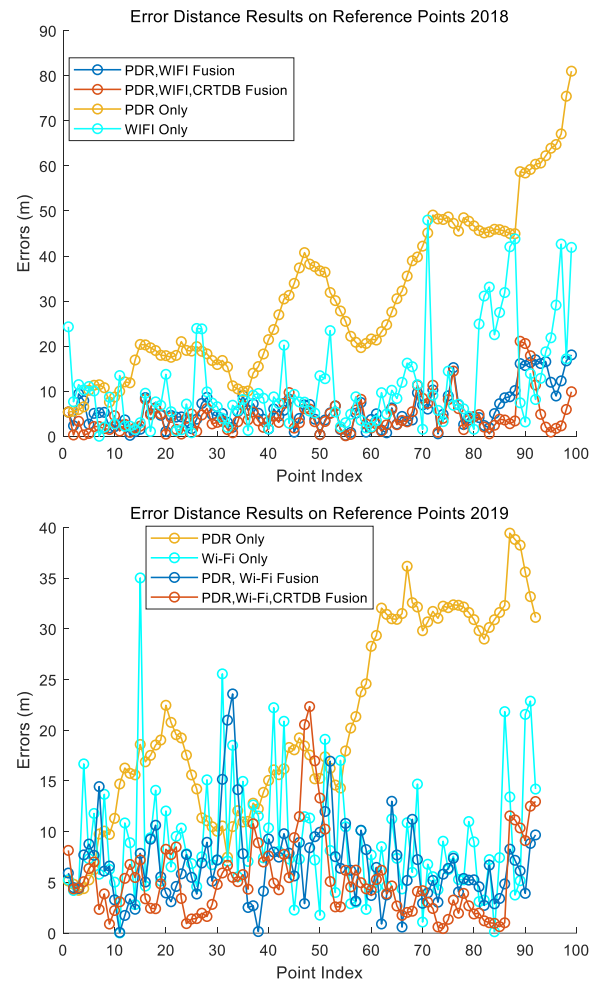


Fig. 9. Error distance comparison at each evaluation point.

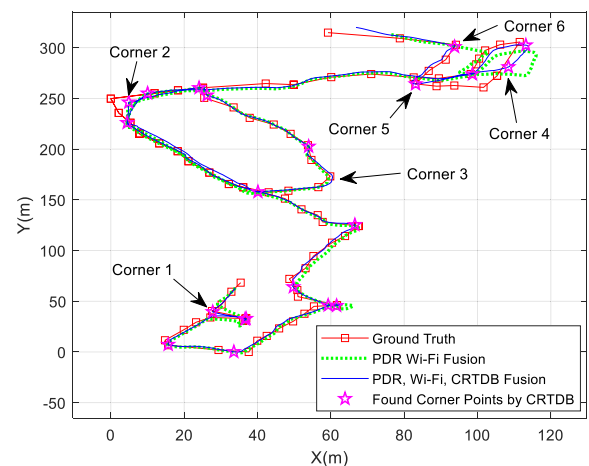


Fig. 10. Experimental results of IPIN2018. To be more intuitive, we translate the geodetic coordinate in degree into local coordinate in meter.

case it can improve the indoor positioning result significantly. Hence it can indeed be used in the real situation. Just like PDR, though it is easily disturbed by abnormal motions, it is still a very important source of positioning. So is CRTDB matching.

As for the computational efficiency of our proposed algorithm, taking the evaluation set of IPIN2018 as an example which contains around 1100 seconds of useful data, it took

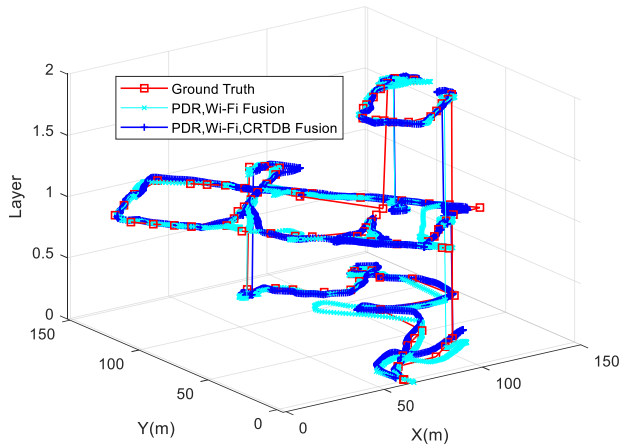


Fig. 11. Experimental results of IPIN2019. To be more intuitive, we translate the geodetic coordinate in degree into local coordinate in meter.

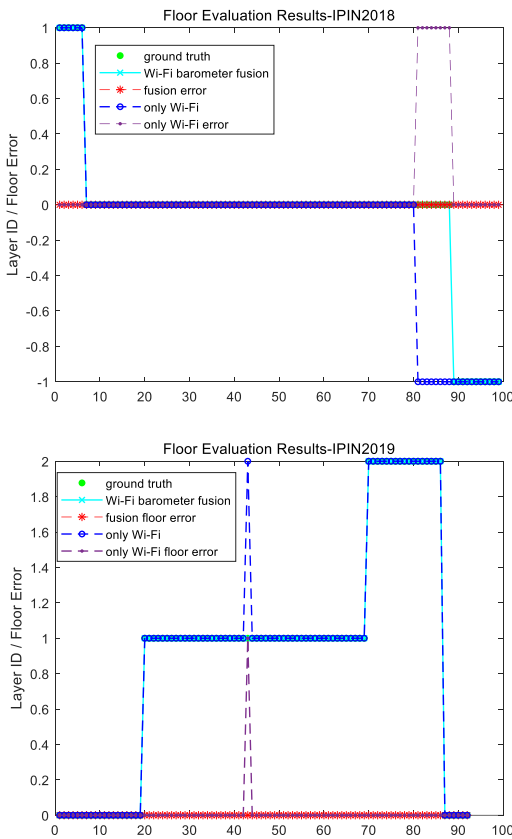


Fig. 12. Layer estimation results.

a total 585.07 seconds for Wi-Fi matching, 0.25 seconds for PDR and Wi-Fi fusion, and 4.18 seconds for CRTDB matching. The average processing time for data per second is 0.54 seconds. The Wi-Fi matching is the most time-consuming part, much larger than the CRTDB matching. This may be due to the fact that the Wi-Fi fingerprint database is much denser than the CRTDB. The total computational efficiency can be improved by two ways for real-time data processing: 1) using C/C++ to implement the whole algorithm because the execution efficiency of MATLAB code is low; 2) using parallel processing to speed up the execution of algorithms.

VIII. CONCLUSION

This paper proposed a positioning system based on Wi-Fi, PDR and CRTDB fusion. We model the step length and heading of the PDR, and propose a KF based trajectory optimization algorithm to optimize the PDR with reference points. Base on the optimized PDR trajectory, we construct Wi-Fi fingerprint database and CRTDB. In this paper, we firstly propose the concept of CRTDB. Then, we propose a CRTDB matching algorithm to obtain the corner reference pseudo-observation. For Wi-Fi and PDR fusion, we propose a screening method to filter out the bad Wi-Fi positioning result. The testing on the two open datasets verifies the effectiveness of our proposed Wi-Fi, PDR and CRTDB combined system.

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